Feynman Path Integrals and the Harmonic Oscillator

Imaginary time, ground-state densities, and the quantum to classical limit

Take home message. In imaginary time, the Feynman path integral for the harmonic oscillator yields a ground-state density that can be compared directly with the Schrödinger solution. On a lattice with N = 7 and T = 4, direct Monte Carlo gives $|\varphi(1)|^2$ = 0.2120 ± 0.0006 (analytic / Schrödinger 0.20755). The Monte Carlo matches the exact lattice answer; the residual offset is short-T physics. The same path sampling then probes $\hbar$ and other potentials.

1. Why path integrals?

Introduction. Feynman sought a formulation of quantum mechanics tied to wave propagation and classical action, rather than an unexplained Schrödinger equation. In one dimension a propagator G advances the wavefunction:

$$\Phi(x_b, t_b) = \int G(x_b, t_b; x_a, t_a)\, \Phi(x_a, t_a)\, dx_a \qquad (1)$$

Like Huygens' principle, earlier amplitudes seed later ones. Physically, G is a sum over all paths connecting a to b, each weighted by the classical action S = ∫(T − V) dt:

$$G(b, a) = \sum_{\text{paths}} e^{\,iS[b,a]/\hbar} \qquad (2)$$

Real time phases oscillate and interfere. The same sum over paths idea is central to lattice QCD, often the only practical non-perturbative tool for the strong force, which motivates studying a clear 1D case first.

2. Imaginary time on a lattice

Making the sum computable. A Wick rotation $\tau = -it$ converts oscillating phases into Boltzmann-like weights $e^{-\varepsilon E}$. Discretising imaginary time, the Euclidean link action becomes

$$S_{\text{link}} = \frac{m(\Delta x)^2}{2\varepsilon\hbar} + \frac{\varepsilon}{\hbar} V\!\left(\frac{x_L + x_R}{2}\right) \qquad (3)$$

At large τ the propagator projects onto the ground state. We study the harmonic oscillator V = ½x² with m = $\hbar$ = 1 on a lattice N = 7, T = 4 (ε = 4/7), whose exact density is

$$|\phi(x)|^2 = \left(\frac{e^{-x^2/2}}{\pi^{1/4}}\right)^2 = \frac{e^{-x^2}}{\sqrt{\pi}} \qquad (4)$$

3. Sampling the path integral

Method. Paths are sampled as free kinetic Brownian bridges pinned at the probe point and reweighted by exp(−Σ εV/$\hbar$); their histogram estimates $|\varphi|^2$. Independent checks: exact HO lattice integral, Metropolis on a ring, and a Schrödinger solver ($E_0 \approx 0.5000$).

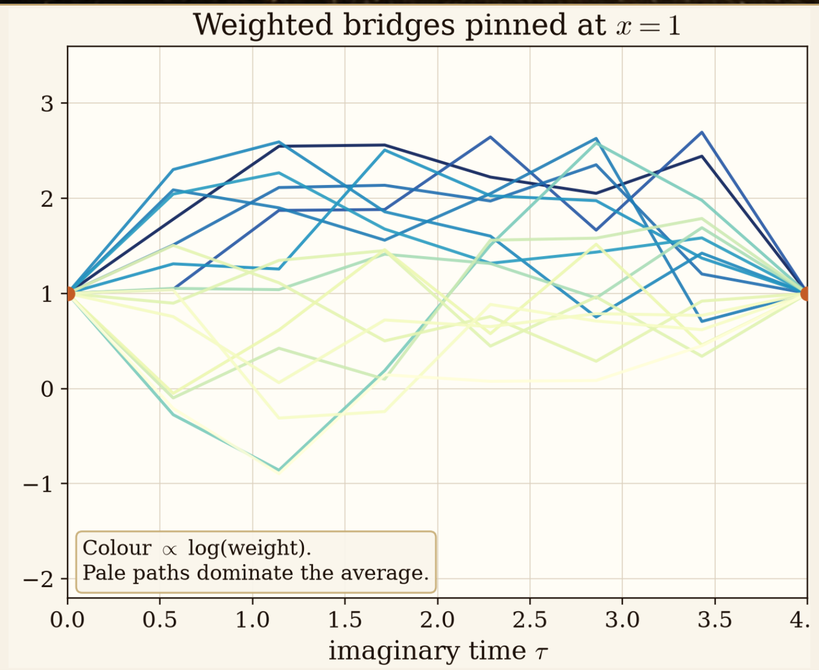


Fig. 0. Bridges pinned at x = 1 (N = 7). Pale paths dominate the weight, the physical idea behind importance sampling.

4. Ground-state density

Results. Comparing path-integral and Schrödinger $|\varphi(x)|^2$ on $x \in [0, 2]$, and quoting the value at x = 1, tests whether the sum over paths recovers ordinary quantum mechanics. Fig. 1 and Table 1 show agreement with the exact lattice; Fig. 2 explains the continuum offset.

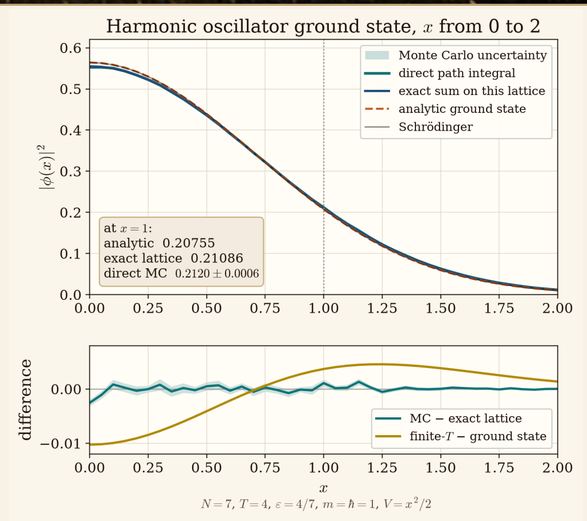


Fig. 1. HO ground-state density (N = 7, T = 4). Direct Monte Carlo tracks the exact lattice; the residual to the continuum ground state is short-T bias (Fig. 2).

Method	$\lvert\varphi(1)\rvert^2$	Notes
Analytic / Schrödinger	0.20755	Eq. (4); continuum GS
Exact lattice (N=7, T=4)	0.21086	same discrete target
Direct MC	**0.2120 ± 0.0006**	weighted bridges
Metropolis (N=7)	0.2110 ± 0.0054	same lattice
Metropolis (N=64, T=12)	0.2087 ± 0.0057	finer production run

Table 1. Probe at x = 1 for the HO ground state.

5. Lattice vs continuum physics

Physical meaning of the offset. Why is 0.2120 above 0.20755 if the sampling is sound? Fig. 2 shows that short imaginary time still mixes excited states: waiting longer (larger T) matters more than refining the number of slices N at fixed short T.

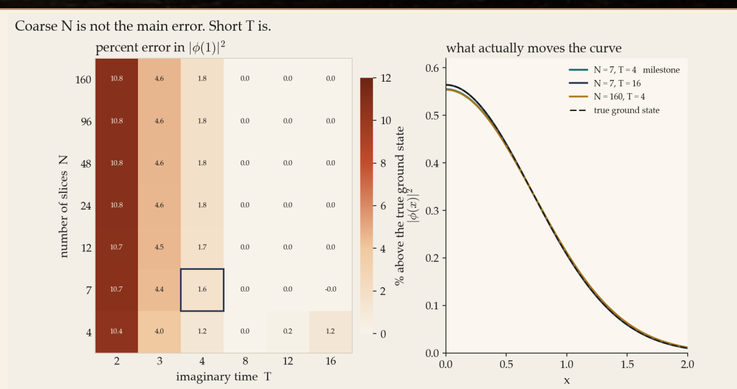


Fig. 2. Error heatmap and densities. Larger T projects onto the true ground state; refining N at T = 4 mainly recovers the exact short-time lattice answer.

6. Paths, $\hbar$, and other potentials

Same path sampling explores fluctuations, the classical limit, and new potentials.

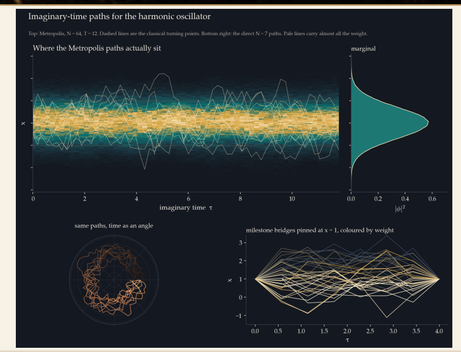


Fig. 3. Worldlines concentrate near x = 0 (N = 64, T = 12), the HO ground-state signature; marginal matches Eq. (4).

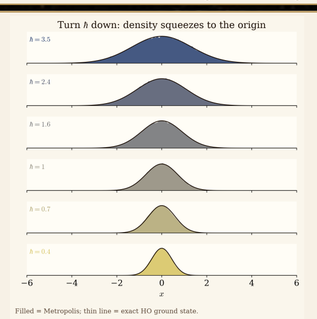


Fig. 4. Quantum to classical crossover: as $\hbar$ falls, $|\varphi|^2$ squeezes to the origin (filled = path integral; thin = exact HO).

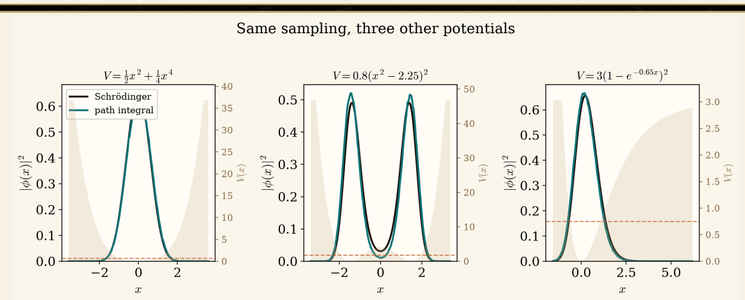


Fig. 5. Anharmonic, double-well, and Morse densities track Schrödinger solutions; local updates still struggle with tunnelling.

7. Outlook and key finding

- HO ground state recovered: $|\varphi(1)|^2 = 0.2120 \pm 0.0006$ on N = 7, T = 4.
- Next: longer T for continuum projection; non-local moves for double-well tunnelling; correlator gaps (≈1.06 vs 1).

Key finding. The path integral matches the exact lattice density (0.21086); the ~2% offset from 0.20755 is short-T physics, not a sampling failure.

References

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